

Solution
HUMAN EYE -06
Class 10 - Science
Section A

1. (c) A is true but R is false.
Explanation: Red colour has longest wavelength.
2. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: After refraction at two parallel faces of a glass slab, a ray of light emerges in a direction parallel to the direction of incidence of white light on the slab. As rays of all colours emerge in same direction, hence there is no dispersion only lateral displacement takes place.
3. (a) Both A and R are true and R is the correct explanation of A.
Explanation: During the daytime, the sky appears blue. This is because the size of the particles in the atmosphere is smaller than the wavelength of visible light, so they scatter the light of shorter wavelengths. The scatter blue light enters our eyes.
4. (d) A is false but R is true.
Explanation: Dispersion of light cannot occur on passing through air contained in a hollow prism. Dispersion takes place because the refractive index of medium for different colours is different.
5. (c) A is true but R is false.
Explanation: A is true but R is false.
6. (a) Both A and R are true and R is the correct explanation of A.
Explanation: In a vacuum speed of light is independent of wavelength, Hence, no dispersion takes place in a vacuum. Thus, a vacuum is a non-dispersive medium in which all colors travel at the same speed.
7. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: The farthest and the nearest point of the eye is based on the distance of the object is placed. Thus, both assertion and reason are true but the reason is not the correct explanation of assertion.
8. (d) A is false but R is true.
Explanation: Magnification produced by mirror,
$$m = \frac{I}{O} = \frac{f}{f-u} = \frac{f}{x}$$

Where, x is distance from focus.
and $m = \frac{\text{Size of image (I)}}{\text{Size of object (O)}}$
9. (a) Both A and R are true and R is the correct explanation of A.
Explanation: $RP \propto \text{diameter of objective}$
10. (c) A is true but R is false.
Explanation: A is true but R is false.
11. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
12. (a) Both A and R are true and R is the correct explanation of A.
Explanation: For thin prism, the angle of prisms A is small. For small A, $D_{\min}$ (minimum deviation) is also small.
$$\text{So, } \mu = \frac{\sin\left(\frac{A+D_{\min}}{2}\right)}{\sin\left(\frac{A}{2}\right)} \dots (i)$$

$$\sin\left(\frac{A+D_{\min}}{2}\right) \approx \frac{A+D_{\min}}{2} \quad (\sin\theta \approx \theta \text{ for small } \theta)$$

$$\text{and } \sin \frac{A}{2} \approx \frac{A}{2}$$

Using above approximation, in equation (i)

$$\mu = \frac{A+D_{\min}}{\frac{A}{2}}$$

$$D_{\min} = (\mu - 1)A$$

Hence, it can be seen that if A is small, $D_{\min}$ is also small.

13. (a) Both A and R are true and R is the correct explanation of A.

Explanation: We know that human eye is the most valuable sense organ because it enables us to see this colourful world. Thus, the given reason is correct explanation of the assertion.

14. (a) Both A and R are true and R is the correct explanation of A.

Explanation: Both A and R are true and R is the correct explanation of A.

15.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: Both A and R are true but R is not the correct explanation of A.

16. (a) Both A and R are true and R is the correct explanation of A.

Explanation: The rainbow is formed when light at the inner surface of the water drop gets internally reflected if the angle between the refracted ray and normal to the drop surface is greater than the critical angle.

17. (a) Both A and R are true and R is the correct explanation of A.

Explanation: Primary rainbow is a result of a three-step process.

i. Refraction at the first surface of a raindrop.

ii. Total internal reflection from the second surface of the raindrop.

iii. Again refraction from the first surface of the raindrop from where the light finally emerges out. The intensity of light is reduced at the second reflection and hence, the secondary rainbow is fainter than the primary rainbow.

18.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: Just after the rain, a large number of small droplets of water remain suspended in the air. Each drop acts like a small prism. When sunlight falls on these drops, the white light splits into seven colours. The dispersed light from a large number of drops forms a continuous band of seven colours.

19.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: Both A and R are true but R is not the correct explanation of A.

20.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: The refractive defects can be corrected by contact lenses and eye lens is composed of fibrous jelly-like material. Thus, both assertion and reason are true but the reason is not the correct explanation of assertion.

21.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: Both A and R are true but R is not the correct explanation of the assertion.

22.

- (d) A is false but R is true.

Explanation: Dispersion takes place because the refractive index of the medium for different wavelengths (colors) is different. The refractive index is inversely proportional to λ by Cauchy's expression as

$$\mu(\lambda) = r + \frac{d}{\lambda^2} + \frac{c}{\lambda^4}$$

Hence, deviation (D) = $(\mu - 1)A$

Since λ_{red} is more than other colors wavelength. So, the deviation is least for red and it appears farthest from the base of the prism.

23.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: Both A and R are true but R is not the correct explanation of A.

24. (a) Both A and R are true and R is the correct explanation of A.

Explanation: We know that the rainbow has seven constituent colours such as Violet, Indigo, Blue, Green, Yellow, Orange and Red. These colours are represented as VIBGYOR. It occurs due to the dispersion of white light i.e., when the white light passes through a glass prism, this phenomenon of splitting white light into VIBGYOR.

25.

- (c) A is true but R is false.

Explanation: Sometimes, the crystalline lens of people at old age becomes milky and cloudy. This condition is called cataract. This causes partial or complete loss of vision. The eyeball is spherical in shape and not elliptical. Thus, the given assertion is true and reason is false.

26.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: As planets are of larger size than stars and much closer to the earth, planets can be considered as a collection of a large number of point-sized sources of light. The total variation in the amount of light entering our eye from all these individual point-sized sources will average out to zero which nullifies the twinkling effect of each other. Therefore, planets do not twinkle.

27.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: Both A and R are true but R is not the correct explanation of A.

28. (a) Both A and R are true and R is the correct explanation of A.

Explanation: Both A and R are true and R is the correct explanation of A.

29. (a) Both A and R are true and R is the correct explanation of A.

Explanation: The continuously changing atmosphere is able to cause variation in the light coming from a point-sized star because of which the star appears to be twinkling.

30.

- (d) A is false but R is true.

Explanation: Higher the refractive index of the prism material, greater is the angle of deviation.

31.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: The light of violet colour deviates most and the light of red colour the least, while passing through a prism. For a prism material refractive index is highest for violet light and lowest for the red light.

32.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: The person cannot see the objects distinctly and comfortably because of accommodation. The vision becomes blurred due to the refractive defects of the eye. Therefore, the given assertion is true, but reason is not the correct explanation of assertion.

33. (a) Both A and R are true and R is the correct explanation of A.

Explanation: We can see the sun and its surroundings appear red colour during both sunrise and sunset. It happens due to the scattering of light is inversely proportional to the fourth power of its wavelength. Thus, both assertion and reason are true and reason is the correct explanation of assertion.

34. (a) Both A and R are true and R is the correct explanation of A.

Explanation: A person with myopia can see nearby objects clearly but cannot see distant objects distinctly. A person with this defect may see clearly up to a distance of a few metres. In a myopic eye, the image of a distant object is formed in front of the retina but not at the retina itself. This defect may arise due to excessive curvature of the eye lens. Thus, both assertion and reason are true and reason is the correct explanation of assertion.

35.

- (d) A is false but R is true.

Explanation: A is false but R is true.

36. (a) Both A and R are true and R is the correct explanation of A.

Explanation: Both A and R are true and R is the correct explanation of A.

37. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
38. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
39. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
40. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
41. (a) Both A and R are true and R is the correct explanation of A.
Explanation: The curvature of the eye lens is modified to change the focal length. When the muscles are relaxed, the lens becomes thin. Therefore, there is an increase in the focal length. This enables us to see distant objects clearly. This modification in lens is done by ciliary muscles. Thus, both assertion and reason are true and reason is the correct explanation of assertion.
42. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Human eye works like camera because of its lens system. Light enters the eye through a thin membrane called the cornea. It forms the transparent bulge on the front surface of the eyeball. The eye lens forms an inverted real image of the object on the retina. Thus, both assertion and reason are true and reason is the correct explanation of assertion.
43. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
44. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.

Section B

45. (a) p, y, z
Explanation: p, y, z
46. (b) $\angle r$, $\angle A$ and $\angle D$
Explanation: $\angle r$, $\angle A$ and $\angle D$
47. (d) $\angle A$ and $\angle i$
Explanation: $\angle A$ and $\angle i$
48. (d) $\angle A$ and $\angle e$
Explanation: $\angle A$ and $\angle e$
49. (a) $\angle 2 + \angle 3 = \angle 1 + \angle 4$
Explanation: Only this option is correct

Section C

50. (b) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)
Explanation:
- **Prism:** It has two triangular bases and three rectangular lateral refracting surfaces. These surfaces are inclined to each other. The angle between its two lateral faces is called 'Angle of Prism'.
 - **Visible spectrum:** The band of seven colours obtained due to the dispersion of white light is called a visible spectrum (VIBGYOR).
 - **Tyndall effect:** It is the phenomenon of scattering of light by the colloidal particles. It can be observed when sunlight passes through a canopy of a dense forest.
 - **Rainbow:** It is formed due to the dispersion and total internal reflection of sunlight by the tiny water droplet, present in the atmosphere. Water droplets act like a prism.

51. **(a)** (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

Explanation: Twinkling of stars is due to atmospheric refraction.

The rainbow is formed due to the dispersion and total internal reflection of sunlight by the tiny water droplet, present in the atmosphere.

The rising sun appears red because the sun is near horizon, and therefore the sunlight has to travel larger distance in the atmosphere which scatters away most of the blue light (shorter wavelength) by the atmospheric particles.

The phenomenon of scattering of light by the colloidal particles is known as the Tyndall Effect. Light of all wavelengths are scattered equally by the clouds and hence clouds appear white.

Section D

52. **(a)** All the colours of the white light move with the same speed

Explanation: All the colors of white light move with the same speed because speed of light doesn't depend on color and has a constant value.